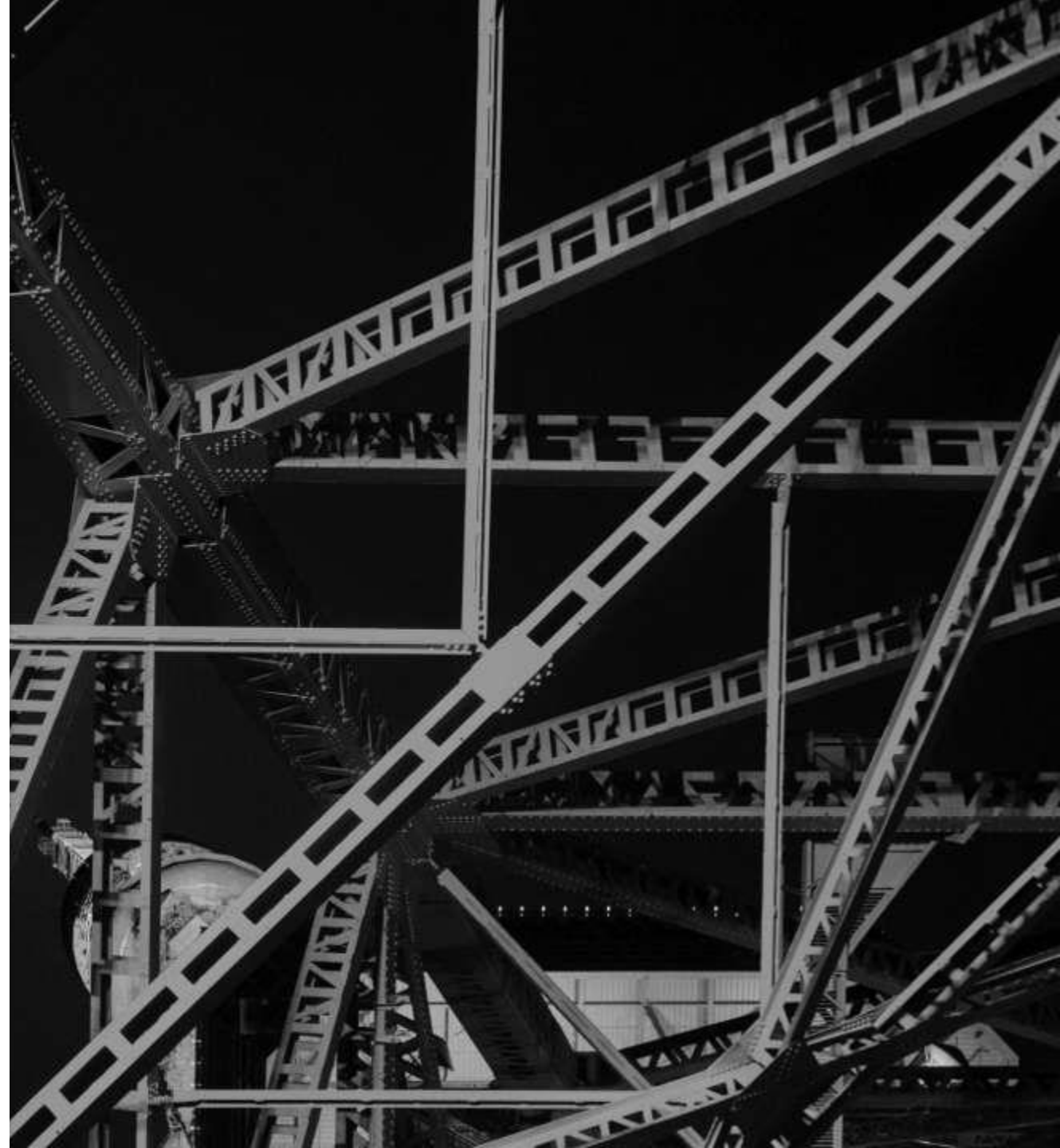




Physical structure

Certification process



Certification Process - Basic Entities

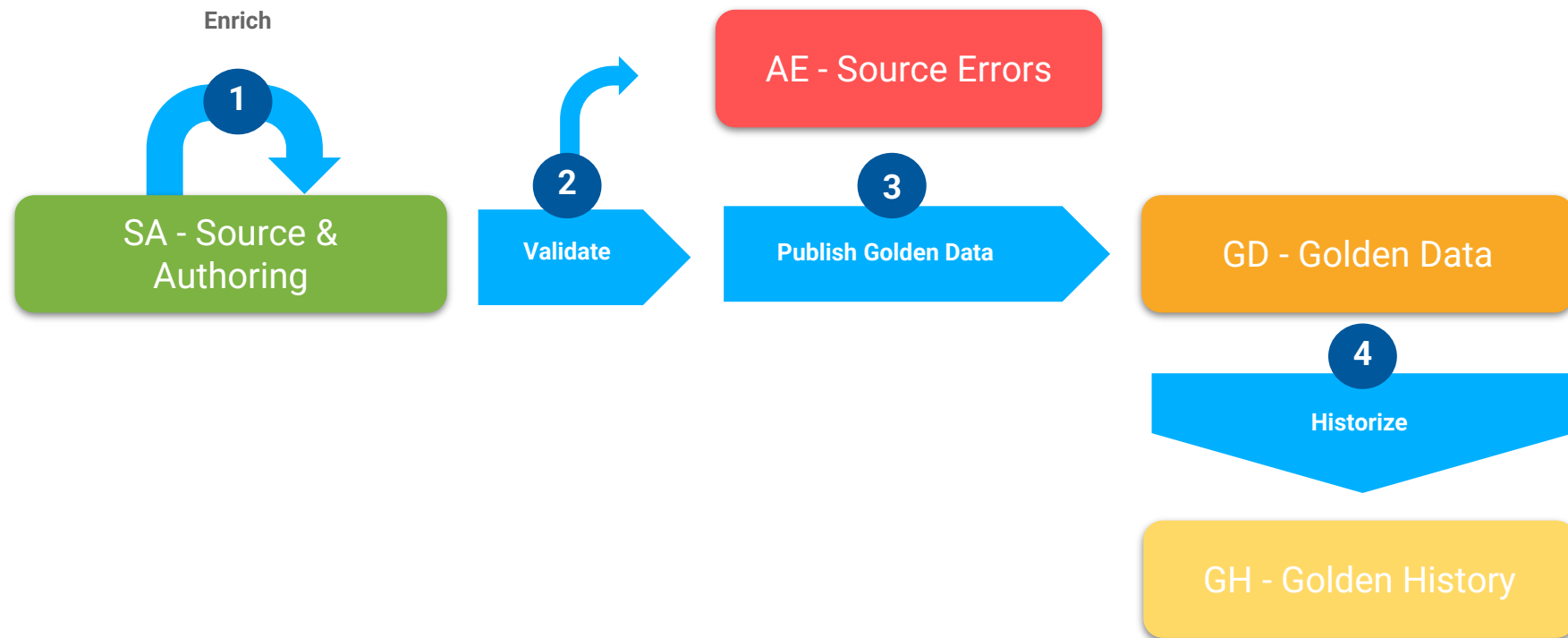


Table structure – Basic vs matched entities

Basics entities don't have a matching process and therefore no consolidation process

Basics entities only have the following tables:

- SA (Source authoring)
- AE (Source authoring errors)
- GD (Golden data)
- **GX (Deleted golden)** – only if '*Delete*' is enabled
- **GH (Golden history)** – only if '*Historization*' is enabled

Certification Process - ID and Fuzzy Matched Entities

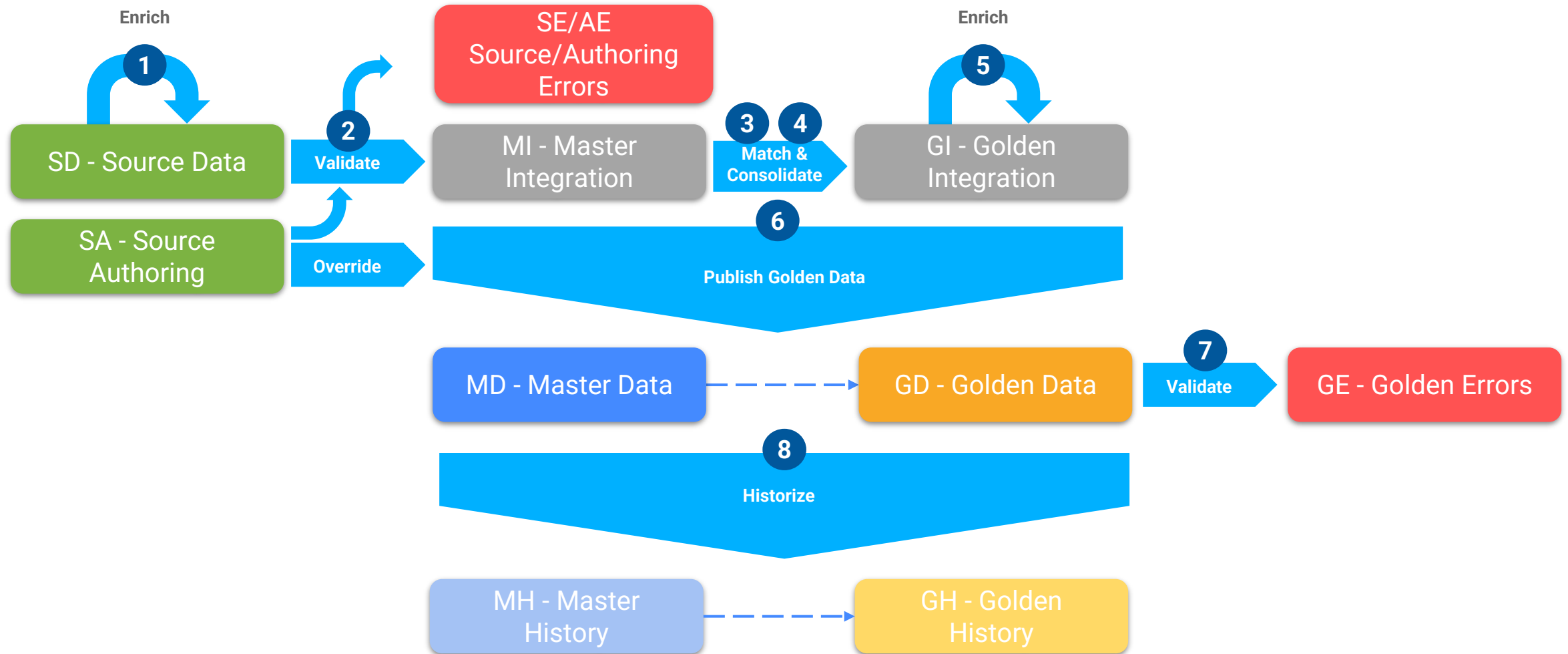


Table structure – ID-matched tables

ID-matched entities have the following tables:

- SA (Source authoring)
- SE (Source errors)
- AE (Source authoring errors)
- **SD (Source data)**
- **MD (Master data)**
- **MI (Master integration)**
- **GI (Golden integration)**
- **GA (Golden authoring)**
- **GD (Golden data)**
- **GE (Golden errors)**

Table structure – Fuzzy-matched tables

Fuzzy-matched entities have the following tables:

- SA (Source authoring)
- SE (Source errors)
- AE (Source authoring errors)
- SD (Source data)
- MD (Master data)
- MI & GI (Master integration and Golden integration)
- GA (Golden authoring)
- GD (Golden data)
- GE (Golden errors)
- MH & GH (Master History and Golden History)
- MX & GX (Master Deletion and Golden Deletion)
- **DU (Duplicates)** – Where matching duplicate pairs are stored
- **UM (Matching groups)** - Stores master records which are being re-grouped by a user in duplicate manager

Tables structures – Basic vs. matching entities

ID and especially **fuzzy-matched entities** go through matching and survivorship. The integration tables support this functionality.

E.g.: **MI** (Master Integration) and **GI** (Golden Integration)

Basic entities skip the integration step. It makes the processing much **faster**.

- They are better suited for reference data, helper tables (like lookups), and join tables for this reason.





Physical structure

Tables structure

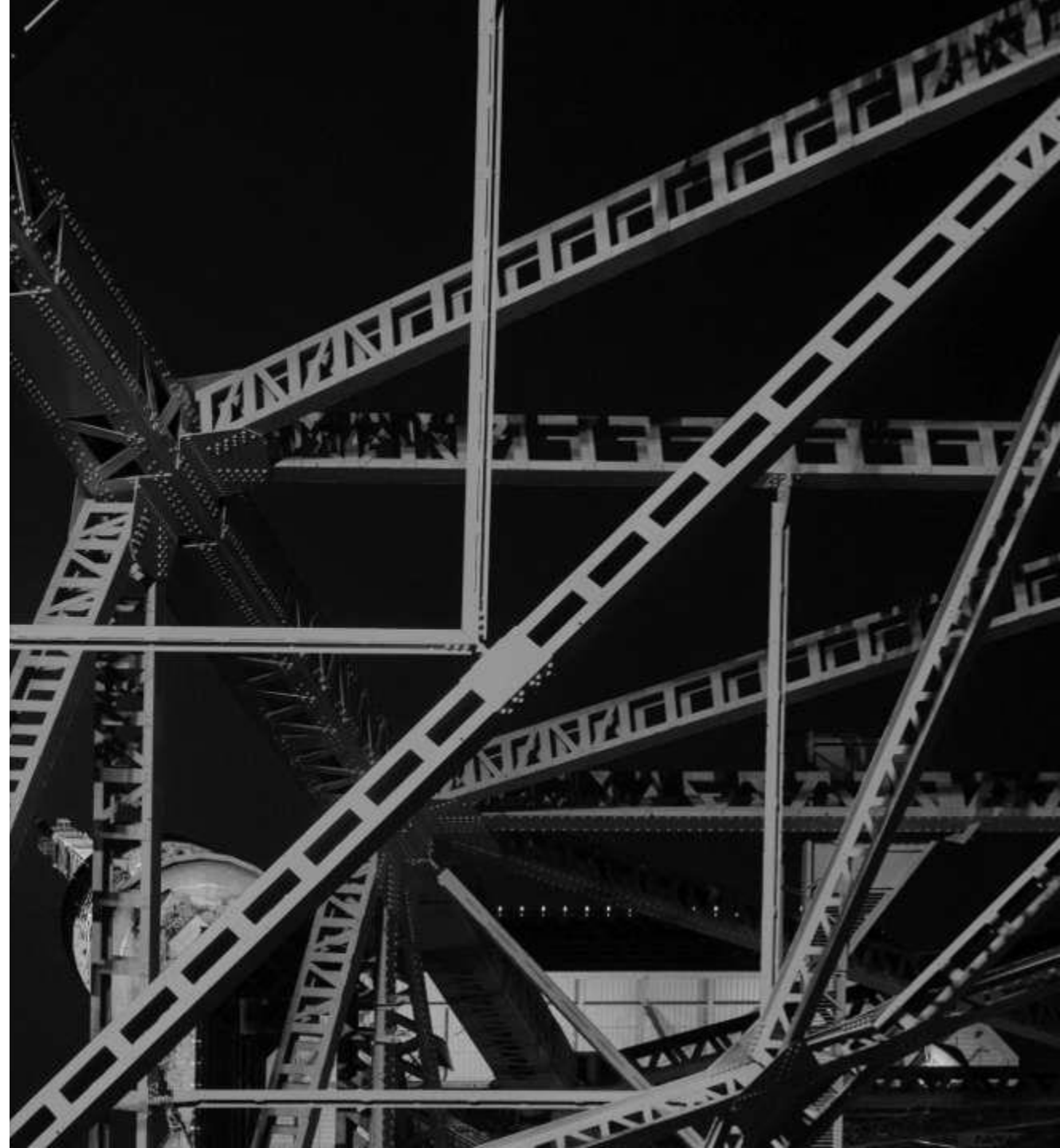


Table structure – Attributes

Simple attributes appear as columns named after their **physical column name**

Complex attributes appear with the pattern:

- <Physical prefix>< Definition attributes physical column name>
- E.g.: *INP**ADDRESS***, *INP**CITY***, *INP**COUNTRY***





Physical structure

Primary keys

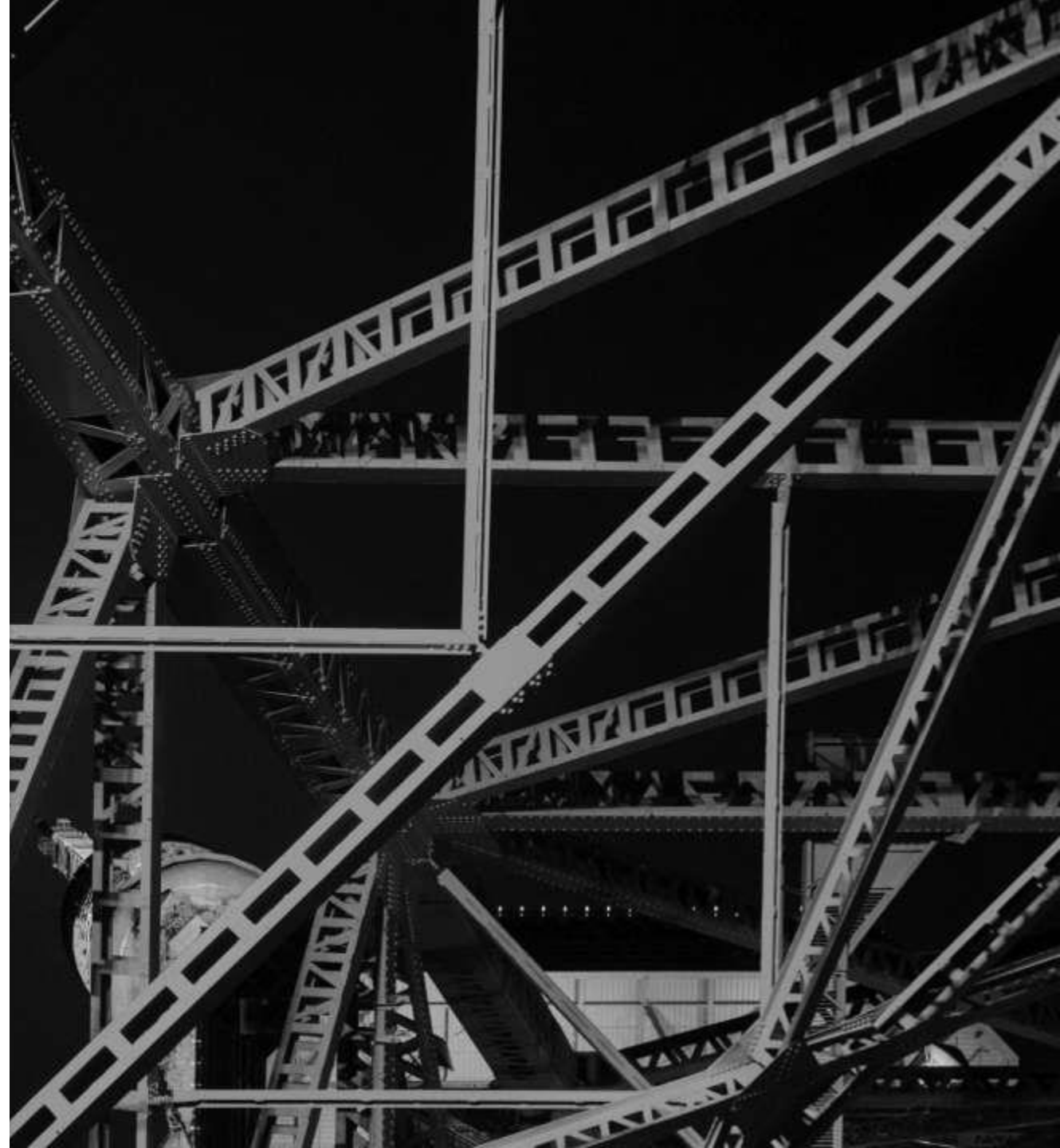


Table structures – Fuzzy vs. ID-matched

ID-matched entities

- A simple column named after the PK attribute's **physical column name**
 - E.g.: PartID, ProductID
- It is a mandatory column loaded in the SD tables
- The value is preserved through the whole process, and matching uses this key

Fuzzy-matched entities:

- A **B_PUBID** column contains the code of the publisher that published the record
- A **B_SOURCEID** column contains the primary key (id) of the source record for this publisher. If it is composite, the columns must be concatenated
- *Golden ID*: A column named after the PK Attribute's **physical column name**
 - The **primary key** value is **generated**
 - It starts to exist in the **master integration** tables



Physical structure

References



Table structure - References

No **foreign keys** (FK) exist at the database level when you deploy the model.

BUT xDM checks references during the **integration process**.

- In the case of a one-to-many relationship, if a child is missing a parent it will be put into the error queue.



Table structure - References

- Data Model
 - Fuzzy-matched parent
 - ID-matched parent
 - Basic entity child

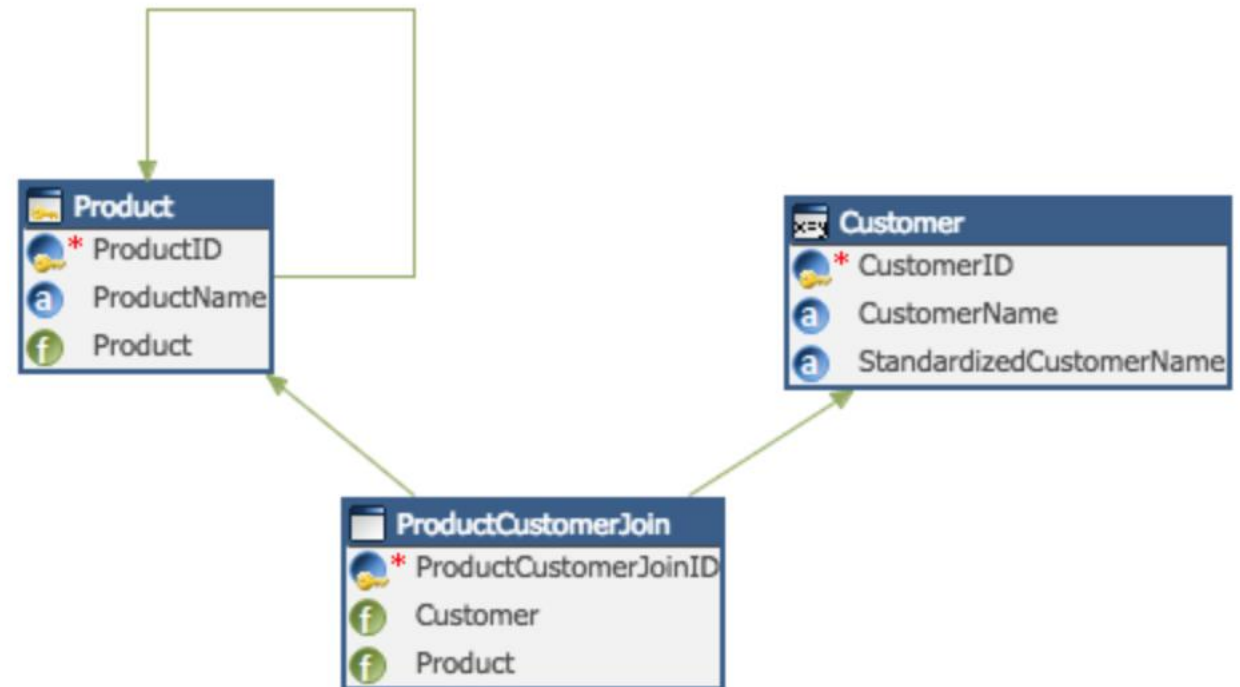


Table structure - References - ID-matched

- References to ID-matched entities
 - Appear as a *F_<Parent Role Name>*
 - E.g.: *F_Product*

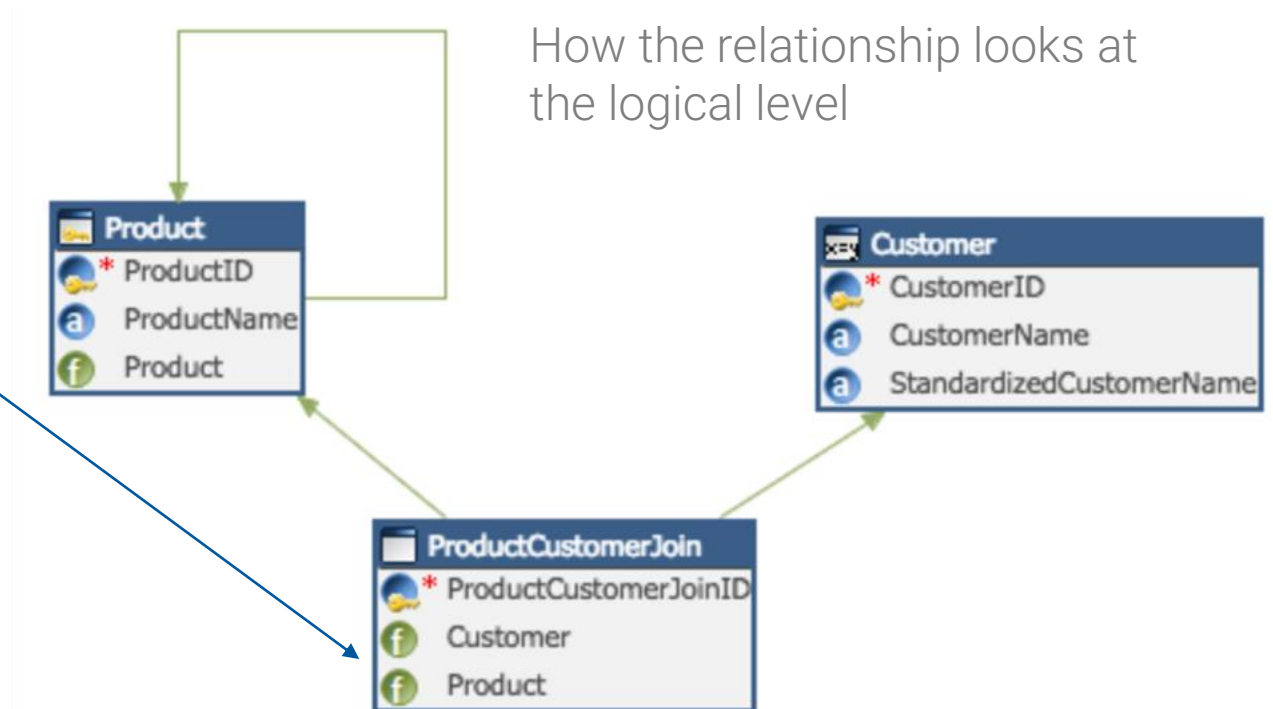
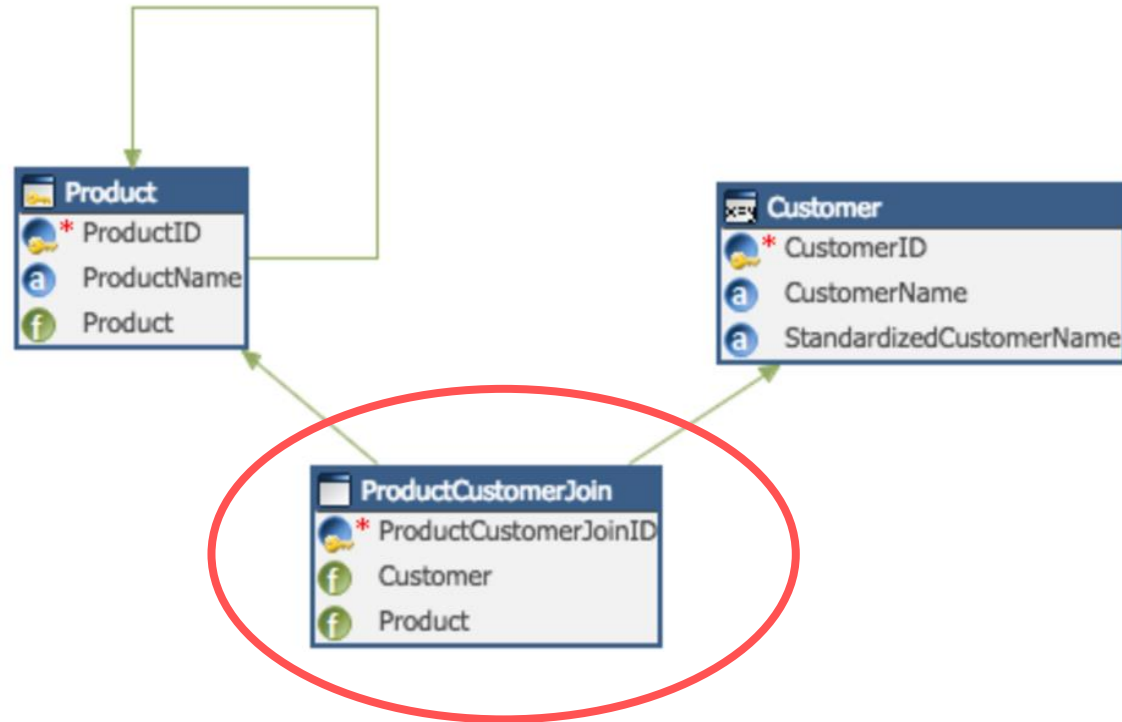


Table structure - References - Fuzzy-matched

References to fuzzy-matched entities

- Referenced source (loaded from the source systems) identified with:
 - *FP_<Physical Name of the Referenced Entity>*: Referenced record publisher
 - *FS_<Physical Name of the Referenced Entity>*: Referenced record ID
 - The referenced entity can be an entity from a different system (source publisher).
E.g.: *FP_CUSTOMER* = 'CRM', *FS_CUSTOMER*='023456'
- Referenced golden identified with:
 - *F_<Physical Name of the Reference To Role Name>*: System generated ID of the referenced golden record.

References - Fuzzy Matching Entities



COLUMN_NAME
PRODUCT_CUSTOMER_JOIN_ID
B_CLASSNAME
B_PUBID
B_SOURCEID
...
B_CREDATE
B_UPDDATE
B_CREATOR
B_UPDATOR
F_CUSTOMER
F_PRODUCT
FP_CUSTOMER
FS_CUSTOMER

Golden customer ID
(fuzzy-matched)

Product ID (ID-
matched)

Source
customer ID
(fuzzy-matched)

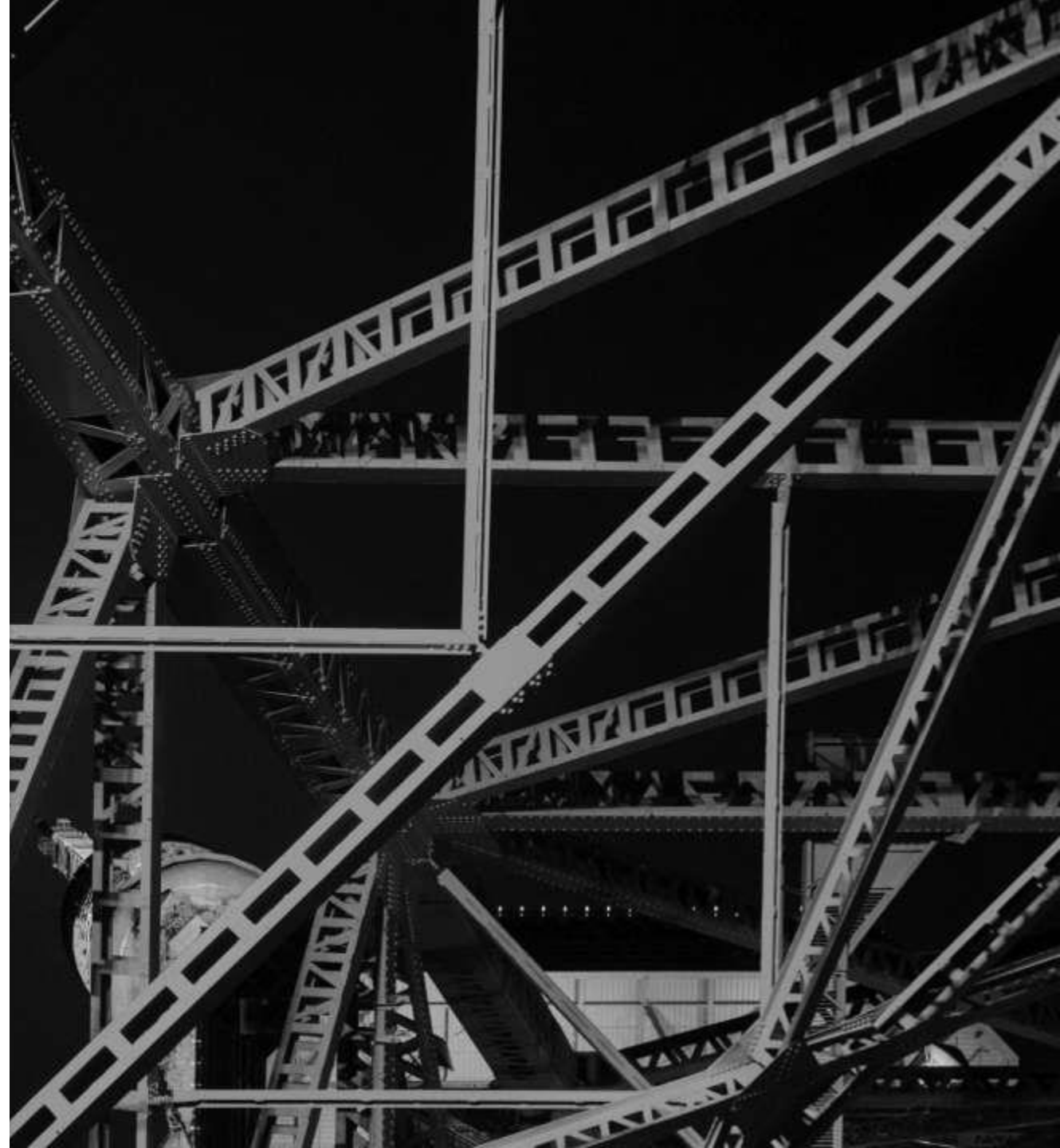
FP_<Parent Role Name> e.g. FP_CUSTOMER = 'CRM'

FS_<Parent Role Name> e.g. FS_CUSTOMER = '023456'



Physical structure

Built-in columns



Built in Columns (1/4)

Column Name	Datatype	Exists in Tables	Description
B_LOADID	NUMBER(38,0)	AE, SA, SD, AE, SE, SF, UM	ID of the load that created this record.
B_BATCHID	NUMBER(38,0)	AE, SE, MI, MD, MX, GA, GF, GI, GE, GD, GP	ID of the batch that created this record.
B_PUBID	VARCHAR2(30 CHAR)	SD, SE, MH, MI, MD, MX, GD, GH, GI, GX, DU, UM	Code of the publisher that published this record.
B_SOURCEID	VARCHAR2(128 CHAR)	SD, SE, MH, MI, MD, MX, GD, GH, GI, GX, DU, UM	ID of the source record in the source publisher system (Fuzzy Matched entities only). See below for a detailed explanation of the primary key options.
B_CLASSNAME	VARCHAR2(128 CHAR)	All	Class name of this record. See Class Name for more information.
B_CONSTRAINTNAME	VARCHAR2(128 CHAR)	AE, SE, GE	Name of the constraint causing this error (Error Records)
B_CONSTRAINTTYPE	VARCHAR2(30 CHAR)	AE, SE, GE	Type of the constraint causing this error (Error Records)
B_ERROR_STATUS	VARCHAR2(30 CHAR)	SA, SD, GD,GH,GX	Flag indicating whether the source of golden record has passed successfully or not validations. Possible values are VALID (if the record has no errors), ERROR (if the record has errors) or OBSOLETE_ERROR (if this record had errors but a newer version of the record fixed them).

Built in Columns (2/4)

Column Name	Datatype	Exists in Tables	Description
B_MATCHGRP	NUMBER(38,0)	MI, MD, MH, MX	ID of the match group for the master record. This column is set when matching takes place.
B_CONFSCORE	NUMBER(38,0)	GH, GI, GD, GX, UM	Confidence score of the golden record. It is the average of the match scores in the match group.
B_XGRP	RAW(16)	MI, MD, MH, MX, UM	Exclusion group ID. An exclusion group represents a group of records for which a user has taken split decisions.
B_HASSUGGMERGE	VARCHAR2(1)	GD, GH, GI, GX, MD, MH, MI, MX, UM	Flag indicating that match and merge suggestions are available for this record.
B_SUGGMERGEID	NUMBER(38,0)	GD, GH, GI, GX, MD, MH, MI, MX, UM	New group ID suggested by the automated matching for this record.
B_SUGGMERGECONFSCORE	NUMBER(38,0)	GD, GH, GI, GX, MD, MH, MI, MX, UM	Confidence score for the suggested group.

Built in Columns (3/4)

Column Name	Datatype	Exists in Tables	Description
B_SUGGMERGEMASTERSCOUNT	NUMBER(38,0)	GD, GH, GI, GX, MD, MH, MI, MX, UM	Number of masters for the suggested group
B_CONFIRMATIONSTATUS	VARCHAR2(30)	GD, GH, GI, GX, MD, MH, MI, MX, UM	Confirmation status for duplicate management: - Confirmed (CONFIRMED): Indicates that a master is confirmed in a golden or that a golden has all its masters confirmed. - Not-confirmed (NOT_CONFIRMED): Indicates that a master is not confirmed or that a golden is entirely made of unconfirmed masters. - Historically confirmed (WAS_CONFIRMED, for master records): If a master was confirmed into to a golden but this golden was fused into another golden. - Partially confirmed (PARTIALLY_CONFIRMED for golden records only): Indicates that a golden has part of his masters confirmed. See the Confirming Golden Record section in the Semarchy xDM Developer's Guide for more information about these statuses.
B_MASTERSCOUNT	NUMBER(38,0)	GH, GI, GD, GX, UM	Number of master records contributing to this golden record.
B_ISCONFIRMED	VARCHAR2(1)	GI, GD, GH, GX	Flag indicated whether this golden record has been validated and the matching confirmed (Fuzzy Matched entities only).

Built in Columns (4/4)

Column Name	Datatype	Exists in Tables	Description
B_ORIGINALBATCHID	NUMBER(38,0)	SA, SD, UM	Batch identifier of the record when it was originally edited out in a stepper.
B_CREDATE	TIMESTAMP(6)	DU, GA, GD, GH, GI, GX, MD, MH, MI, MX, SA, SD, UM	Creation date of the record.
B_UPDDATE	TIMESTAMP(6)	DU, GA, GD, GH, GI, GX, MD, MH, MI, MX, SA, SD, UM	Last update date of the record.
B_CREATOR	VARCHAR2(128 CHAR)	DU, GA, GD, GH, GI, GX, MD, MH, MI, MX, SA, SD, UM	Creator of the record.
B_UPDATOR	VARCHAR2(128 CHAR)	DU, GA, GD, GH, GI, GX, MD, MH, MI, MX, SA, SD, UM	Last updater of this record.



For more details: https://www.semarchy.com/doc/semarchy-xdm/xdm/5.3/Integrate/table-structures.html#_built_in_columns